

Module Guide

Foundations of Computer Systems COMP4009

Level 4 15 Credits
Module Leader: Shumao Ou
2026-27

Purpose of this guide

The purpose of this Module Guide is to provide you with specific information that under-pins the design, delivery and management of this module. It is your responsibility to familiarise yourself with the contents of this Module Guide and to talk to your seminar tutor if you have any questions.

This isn't the only guide available to you. Alongside this guide you will also have access to the University Handbook which will provide you with information that is central to your studies, including policies and regulations, student support and wellbeing and all the services available to you through Student Support.

You will also be given a Programme Handbook which will detail the design, delivery principles and management of your programme. It will also provide you with an overview of your key contacts for your programme of study and list all the modules you will be / might be studying throughout the duration of your award.

Contents

Module Guide	0
Purpose of this guide	1
Contents	2
Module introduction	3
Module leader contact details	3
Time Slots and Rooms	3
Library Support.....	3
Changes made to the module (if any) in response to student and other feedback .	4
Module Study Plan.....	5
Module syllabus	6
Recommended reading list	6
Assessment information	7
Artificial Intelligence and its use in this assessment.....	7
ISP Extensions	8
Marking and moderation of your work	8
Feedback	8

Module introduction

Module leader contact details

Name: Dr Shumao Ou
Room: NHHB.1.01, Headington Campus
Email: sou@brookes.ac.uk

Time Slots and Rooms

Lectures: Tuesday 13:00-15:00, JHB.B.08
(Lecturer: Shumao Ou & Safras Iqbal)
Practical Sessions: Tuesday 15:00-18:00 NHHB.1.18 & NHHB.1.17

Library Support

Contact the Library for support with resources and referencing for this module:
email libraryenquiries@brookes.ac.uk, use our Live Chat service (yellow button on any Library Web page) or go to the [Course Resource Help page for your subject](#).

Changes made to the module (if any) in response to student and other feedback

The two pieces of coursework are changed to timed assessment.

Module Study Plan

Lecture/Practical schedule (tentative):

Week	Topic	Activity	Assessment
1	Numbering Systems	Lecture & Practical lab	
2	Data Representation 1	Lecture & Practical lab	
3	Data Representation 2	Lecture & Practical lab	
4	Computing Logic	Lecture & Practical lab	
5	Boolean Expression	Lecture & Practical lab	
6	<i>Consolidation</i>	Coursework 1	Timed assessment (2 hours, 40%)
7	Assembly Language	Lecture & Practical lab	
8	Parallelisation	Lecture & Practical lab	
9	OS File Systems OS Sharing	Lecture & Practical lab	
10	OS Memory	Lecture & Practical lab	
11	Assembly Programming	Lecture & Practical lab	
12	<i>Consolidation</i>	Coursework 2	Timed assessment (3 hours, 60%)

Module syllabus

- Understand modern computer architecture
- Apply architecture concepts to software design
- Gain insight in computer development
- Understand functional role of computers in society
- Design logical computer components
- Understand modern programming language design decisions in the context of the computer architecture

Recommended reading list

Dale, N. "Computer Science Illuminated" 5th Edition. Jones and Bartlett, 2012. ISBN: 978- 1449672843

Forouzan & Mousharaf. "Foundations of Computer Science" 2nd edition. Thomson Learning. ISBNL: 9781844807000

G.Knott & N. Waites. *Computing 3rd edition*, Business Education Publisher Limited. ISBN 190188215

J.G. Brookshear. *Computer Science an overview, 10th Edition*, Addison Wesley.

Maths book that can be useful

Lipschutz, S. Schauman's Outline of Essential Computer Mathematics, 1982

Interesting book that can be useful to understand how computer works and the philosophy behind it

Petzold, C., Code The hidden language of Computer Hardware and Software

Assessment information

This module follows the principles of the University's Assessment and feedback policy developed in conjunction with the Student Union, to ensure good practice and transparency in assessment and feedback processes. The Assessment and feedback policy can be found in your Programme Handbook or on your Programme's Brookes Virtual site as well as on the University webpage of A-Z of Policies.

Please note: the Institutional University Handbook which will provide you with information that is central to your studies, including policies and regulations, student support and wellbeing and all the services available to you through Student Support. The core information is also available on Moodle via the drop down menu under 'Student Help'

This module is assessed by **100%** coursework.

The assessment for this course is divided into two Timed Assessment coursework:

Coursework 1 - Design a Logic Circuit, Report, **40%**

Coursework 2 – Design of Operating Systems, **60%**

Coursework 1 - Design a Logic Circuit

You will design a logic circuit. It is times assessment. You will do this work in 2 hours under supervision.

Assignment length

There is no page or word limitation. You need to finish the tasks required. It is advised to read the coursework script carefully and to address all the marking criteria properly.

Coursework 2 – Design of Operating Systems

You will answer questions about the design considerations of computer Operating Systems (OS). It is times assessment. You will do this work in 3 hours under supervision.

Artificial Intelligence and its use in this assessment

All students should complete the 'Use of Artificial Intelligence' course available on Moodle. This link will take you to further guidance on [the use of AI tools at Brookes](#). This link will take you to the Moodle course [Use of Artificial Intelligence](#).

For this assessment specifically, the extent you can use AI tools is detailed below,

- If you use any AI tools for this assessment, you need to state your use in the [declaration form](#) when you submit your work.
- You must **not** use AI tools for this assessment.
- In this assignment, you can use AI tools for preparation.

ISP Extensions

If you have an Inclusive Support Plan you can check whether you have coursework extensions and how many weeks on your Student Information page. Contact your module leader if you are unclear about your deadlines.

Do not confuse coursework extensions agreed as part of an Inclusive Support Plan (ISP) with Exceptional Circumstances. There are no limits to how often you can use the extensions agreed as part of your ISP.

There are final submission dates which no extensions can go beyond. These are the same for extensions agreed through the Exceptional Circumstances Procedure or an ISP. You can find this on the [Exceptional Circumstances website](#). If you are not sure whether your ISP coursework extension would go past the final hand-in date, always contact your Module Leader to confirm your deadline.

Marking and moderation of your work

Following internal moderation, a sample of work is reviewed by the External Examiner for the programme to ensure that the standards applied are comparable to those at other institutions. To read how your work is moderated please go to your Programme Handbook for details.

Feedback

Feedback on your work will be provided in a range of ways at various times throughout this module, and different feedback will serve slightly different purposes. Feedback is designed to support your learning and help you to improve subsequent work, so you need to engage and get the most out of the feedback provided.

Please note that feedback is provided throughout the module, not just on formal tasks. It will be provided on your work and contribution in class, on the formal assessment tasks and, in some circumstances, during staff office hours.

If you would like further information about feedback, or how to use it, please talk to your academic tutor on this module or your Academic Adviser or Programme Lead.

Assessment feedback will normally be provided within **15** working days of submission.

Please note that all marks are provisional until they are ratified by an Examination Committee.